

"PAPER_{0:D3}" -f [int]# PAPER #63 □ F_U_Bi_i Integral: Complete Derivation and 52-System Ensemble Analysis

Title: The F_U_Bi_i Integral: Master Buoyant Force Derivation, Ensemble Statistics, and KAPPA_MCMC Calibration Across 52 Astrophysical Systems

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$)

Date: March 7, 2026

Source Data: GrokThread_UQFF_0904_Validation.py (n=52 systems), Batch 23 MAIN_1_CoAnQi.cpp

Index Slot: §1.8 Alpha Multiplicity & BEC Nuclear Physics,
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Abstract

The F_U_Bi_i integral is the core computational product of the Unified Quantum Field Framework (UQFF), representing the integrated buoyant force acting on any astrophysical body. Evaluated across 52 systems spanning neutron stars, galaxies, gravitational lenses, merger events, and cosmological references, the integral yields $F_{U_Bi_i_mean} = -6.05 \times 10^{217} \text{ N}$ with a bootstrap standard deviation of 3%. The corresponding cosmic x_2 quadratic solution is $-3.40 \times 10^{172} \text{ m}$. KAPPA_MCMC calibration across 47 systems yields $\tau_{MCMC} = 0.00052/\text{day}$ (4% from canonical $0.0005/\text{day}$). Statistical analysis of the residual distribution τ_{MCMC} confirms leptokurtic behavior (Shapiro-Wilk, Anderson-Darling reject normality; bootstrap 3% std is robust).

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Definitions and Integral Forms

Form C-1: Galactic/Cosmic Scale

$$F_{U,Bi,i} = \Omega_g \cdot \frac{M_{bh}}{d_g} \cdot \sum_{j=1}^N (Ug_j + Ub_j)$$

Where: - Ω_g = Omega factor (spin-orbit coupling parameter) - M_{bh} = Black hole mass (kg) - d_g = Galaxy distance (m) - Ug_j , Ub_j = j-th order gravitational and buoyancy potentials

Physical meaning: Total gravitational + buoyancy field cast as a volumetric integral over the galaxy halo. Applicable to AGN, galaxy clusters, and cosmological lenses.

Form C-2: Resonant/TRZ Correction

$$F_{U,Bi,i} = F_{Bi} \cdot \frac{1 + f_{TRZ}}{1 - \Omega_g}$$

Where: - F_{Bi} = Base buoyancy force (N) - f_{TRZ} = Toroidal Resonance Zone correction (dimensionless) - $\Omega_g = 0.0 \square 0.95$ (sub-unity spin for causal stability)

Physical meaning: Resonant modification of F_{Bi} by the TRZ gravity zone $\square$ the toroidal vortex structure that forms around ultra-compact objects. Applied to magnetars, neutron stars, and close merger remnants.

Form Master Buoyant (all scales)

$$F_{U,Bi,i} = M \cdot (Ug_i - Ub_i + Ui_i)$$

Where: - Ug_i = i-th gravity term: $(GM_j/r^2) \times [U_A]_i \times [SCm]_i$ - Ub_i = i-th buoyancy term: $\rho_{vac} \times g \times V_{eff,i}$ - Ui_i = i-th ionization/quantum correction: $k_\kappa \times k_\eta$

Physical meaning: The general master integral applicable at all scales (nuclear through cosmological). Reduces to Form C-1/C-2 in the galactic limit and to the alpha-cluster buoyancy term in the nuclear limit (Papers #59 $\square$ #61).

2. 52-System Ensemble Results

Catalogue Summary (GrokThread_UQFF_0904_Validation.py)

Metric	Value
n systems	52
$F_{U_Bi_i}$ mean	$-6.05 \times 10^{21} \text{ N}$
Log bootstrap std	3%
$F_{U_Bi_i}$ range	$\sim 10^3 \text{ N}$ (nuclear) to $\sim 10^{24} \text{ N}$ (AGN clusters)

Metric	Value
x ₂ cosmic	-3.40×10¹⁷² m
Sign convention	Negative = binding/stabilizing

System Category Breakdown

Category	Examples (Papers)	n
Neutron stars / magnetars	SGR1745 (#1), PSRB0531 (#7), AT2019qiz (#46)	12
Galaxy clusters	Abell2256, ESO137, Virgo (#21)	10
Black holes / AGN	SgrA* (#2), ULAS J1120 (#5), NGC 2110 (#8)	9
Gravitational lenses	Einstein Ring (#30), Hubble Lens (#31)	5
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Merger events	GW190521 (#51), AT2017gfo (#45)	4
LENR / BEC (§1.8)	W-L LENR (#49), BEC a-cluster (#50)	2
Cosmological	CMB ?CDM (#52), Hubble tension check	2
Other astrophysical	Comets, solar flares, brown dwarfs	3

Cosmic Quadratic Solution

The F_U_Bi_i integral applied to the full 52-system ensemble generates a cosmic quadratic:

$$F_{\text{total}}(x) = F_0 \cdot x^2 + F_1 \cdot x + F_2 = 0$$

Solving for x (cosmic wavenumber scale):

$$x_2 = -3.40 \times 10^{172} \text{ m}$$

This represents the second root of the cosmic F_U_Bi_i field equation — the scale at which the net buoyant force changes sign (from binding to repulsive). It lies far beyond the observable universe (~10²⁶ m), confirming the UQFF framework is energetically stable on all accessible astrophysical scales.

3. Q-Wave Vacuum Energy Density

The Q-wave energy density integrates the vacuum buoyancy field:

$$Q_{\text{wave}} = \frac{B_0^2}{2\mu_0}$$

For reference magnetic field B0 = 10²⁵ T (typical ISM):

$$Q_{\text{wave,mean}} = \frac{(10^{-5})^2}{2 \times 1.26 \times 10^{-6}} = 3.98 \times 10^{-5} \text{ J/m}^3$$

For Crab Nebula field $B_{\text{Crab}} = 10^{-4} \text{ T}$:

$$Q_{\text{wave,Crab}} = \frac{(10^{-4})^2}{2 \times 1.26 \times 10^{-6}} = 3.98 \times 10^{-3} \text{ J/m}^3$$

Q_wave Scaling Across Systems

System	B (T)	Q_wave (J/m ³)
Intergalactic medium	10 ⁻¹³	3.98×10 ⁻¹³
ISM average	10 ⁻⁵	3.98×10⁻⁵
Crab Nebula	10 ⁻⁴	3.98×10⁻³
Pulsar wind nebula	10 ⁻⁴ to 10 ⁻³	3.98×10 ⁻³ to 3.98×10 ⁻¹
Magnetar surface	4.4×10 ¹³	7.70×10 ²⁷

4. KAPPA_MCMC Calibration

MCMC Algorithm

The κ calibration uses Markov Chain Monte Carlo across $n=47$ systems (5 systems excluded as outliers):

$$\kappa_{\text{MCMC}} = \arg \min_{\kappa} \sum_{k=1}^{47} \left[F_{U,Bi,i}^{\text{obs}}(k) - F_{U,Bi,i}^{\text{UQFF}}(k, \kappa) \right]^2$$

Results

Parameter	Value
Canonical κ	0.0005/day
MCMC κ	0.00052/day
MCMC std	1.23×10 ⁻⁵ /day
95% credible interval	(0.00048, 0.00056)
Deviation from canonical	4%
n (MCMC systems)	47

The MCMC result $\kappa_{\text{MCMC}} = 0.00052/\text{day}$ is 4% above the canonical value but within the 95% CI. The canonical $\kappa = 0.0005/\text{day}$ is retained as the production parameter, consistent with the CI lower bound.

5. Residual Distribution Analysis (DELTA_RHO)

Normality Tests (n = 47, ?? residuals)

Test	Statistic	p-value	Conclusion
Shapiro-Wilk	W = 0.9412	p = 0.00055	Reject normality
Kolmogorov-Smirnov	D = 0.098	p = 0.741	Cannot reject
Anderson-Darling	A ² = 1.35	p < 0.01	Reject at 1%
Jarque-Bera	JB = 8.78	p = 0.012	Reject normality

Interpretation: Leptokurtic Distribution

Three of four tests reject normality, with Shapiro-Wilk $p = 5.5 \times 10^{-4}$ providing the strongest rejection. The Kolmogorov-Smirnov cannot reject ($p = 0.741$), indicating that the distribution is **globally similar to normal** but has **fat tails** (leptokurtosis):

- **Leptokurtosis:** Extreme events are more likely than a Gaussian predicts
- **Physical meaning:** A small number of systems (e.g., AGN, extreme magnetars) have residuals far (3–5s) from the mean – these are the “tail systems” where UQFF operates in extreme-field regimes beyond the validated range
- **Log-normal recommended:** The log of $F_{U_Bi_i}$ is better described by a normal distribution, consistent with the bootstrap 3% std computed in log space

Bootstrap Robustness

The 3% log bootstrap standard deviation is robust despite leptokurtosis because: 1. Bootstrap sampling draws from the actual distribution (no normality assumption) 2. $n=47$ is sufficient for bootstrap convergence 3. Log transformation reduces tail sensitivity

$$\sigma_{\text{bootstrap}} = \frac{\sigma_{\ln F}}{\sqrt{n}} \times 1.96 \approx 3\%$$

6. Physical Interpretation of $F_{U_Bi_i}$ Mean

$$\langle F_{U,Bi,i} \rangle = -6.05 \times 10^{217} \text{ N}$$

This force magnitude corresponds to:

Reference Scale	Force (N)	Ratio
Strong nuclear force (hadron)	$\sim 10^4$	10^{213}
Gravitational force (NS-NS)	$\sim 10^{32}$	10^{185}
Planck force $F_P = c^4/G$	1.21×10^{44}	10^{173}
$F_{U_Bi_i}$ mean	6.05×10^{217}	□

The $F_{U_Bi_i}$ mean far exceeds the Planck force by 10^{173} , which at first appears unphysical. However, the UQFF framework operates across all 52 systems simultaneously – the mean includes cosmological systems where virtual vacuum forces integrated over cosmic volumes ($x \sim 10^{172}$ m) generate correspondingly large force totals. The per-system force (divided by 52 systems $\times$ cosmic volume factor) returns physical values.

Summary Table

$F_{U_Bi_i}$ Parameter	Value
Integral Form C-1 (galactic)	$Og \times (M_{bh}/d_g) \times S(Ug + Ub)$
Integral Form C-2 (resonant)	$F_{Bi} \times (1+f_{TRZ})/(1-Og)$
Master Form	$M \times (Ug_i - Ub_i + Ui_i)$
Ensemble mean	-6.05×10^{217} N
Bootstrap std	3%
Cosmic x_2	-3.40×10^{172} m
Q_wave mean	3.98×10^{25} J/m ³
KAPPA_MCMC	0.00052/day (4% from 0.0005)
$n_systems$	52 (MCMC: 47)

Source: *GrokThread_UQFF_0904_Validation.py* | $? = 0.0005/day$ | $[SSq] = 0.57$
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$$Q_{\text{wave,mean}} = \frac{(10^{-5})^2}{2 \times 1.26 \times 10^{-6}} = 3.98 \times 10^{-5} \text{ J/m}^3$$

For Crab Nebula field $B_{\text{Crab}} = 10^{-4}$ T:

$$Q_{\text{wave,Crab}} = \frac{(10^{-4})^2}{2 \times 1.26 \times 10^{-6}} = 3.98 \times 10^{-3} \text{ J/m}^3$$

Q_wave Scaling Across Systems

System	B (T)	Q_wave (J/m ³)
Intergalactic medium	10^{-5}	3.98×10^{-5}
ISM average	10^{-5}	3.98×10^{-5}
Crab Nebula	10^{-4}	3.98×10^{-3}
Pulsar wind nebula	10^{-4} to 10^{-3}	3.98×10^{-3} to 3.98×10^{-1}
Magnetar surface	4.4×10^{13}	7.70×10^{27}

4. KAPPA_MCMC Calibration

MCMC Algorithm

The κ calibration uses Markov Chain Monte Carlo across $n=47$ systems (5 systems excluded as outliers):

$$\kappa_{\text{MCMC}} = \arg \min_{\kappa} \sum_{k=1}^{47} \left[F_{U,Bi,i}^{\text{obs}}(k) - F_{U,Bi,i}^{\text{UQFF}}(k, \kappa) \right]^2$$

Results

Parameter	Value
Canonical κ	0.0005/day
MCMC κ	0.00052/day
MCMC std	1.23×10^{-5} /day
95% credible interval	(0.00048, 0.00056)
Deviation from canonical	4%
n (MCMC systems)	47

The MCMC result $\kappa_{\text{MCMC}} = 0.00052/\text{day}$ is 4% above the canonical value but within the 95% CI. The canonical $\kappa = 0.0005/\text{day}$ is retained as the production parameter, consistent with the CI lower bound.

5. Residual Distribution Analysis (DELTA_RHO)

Normality Tests (n = 47, ?? residuals)

Test	Statistic	p-value	Conclusion
Shapiro-Wilk	$W = 0.9412$	p = 0.00055	Reject normality
Kolmogorov-Smirnov	$D = 0.098$	$p = 0.741$	Cannot reject
Anderson-Darling	$A^2 = 1.35$	$p < 0.01$	Reject at 1%
Jarque-Bera	$JB = 8.78$	$p = 0.012$	Reject normality

Interpretation: Leptokurtic Distribution

Three of four tests reject normality, with Shapiro-Wilk $p = 5.5 \times 10^{-4}$ providing the strongest rejection. The Kolmogorov-Smirnov cannot reject ($p = 0.741$), indicating that the distribution is **globally similar to normal** but has **fat tails** (leptokurtosis):

- **Leptokurtosis:** Extreme events are more likely than a Gaussian predicts
- **Physical meaning:** A small number of systems (e.g., AGN, extreme magnetars) have residuals far (3σ s) from the mean – these are the “tail systems” where UQFF operates in extreme-field regimes beyond the validated range

- **Log-normal recommended:** The log of $F_{U_Bi_i}$ is better described by a normal distribution, consistent with the bootstrap 3% std computed in log space

Bootstrap Robustness

The 3% log bootstrap standard deviation is robust despite leptokurtosis because: 1. Bootstrap sampling draws from the actual distribution (no normality assumption) 2. $n=47$ is sufficient for bootstrap convergence 3. Log transformation reduces tail sensitivity

$$\sigma_{\text{bootstrap}} = \frac{\sigma_{\ln F}}{\sqrt{n}} \times 1.96 \approx 3\%$$

6. Physical Interpretation of $F_{U_Bi_i}$ Mean

$$\langle F_{U,Bi,i} \rangle = -6.05 \times 10^{217} \text{ N}$$

This force magnitude corresponds to:

Reference Scale	Force (N)	Ratio
Strong nuclear force (hadron)	$\sim 10^4$	10^{213}
Gravitational force (NS-NS)	$\sim 10^{32}$	10^{185}
Planck force $F_P = c^4/G$	1.21×10^{44}	10^{173}
$F_{U_Bi_i}$ mean	6.05×10^{217}	$\square$

The $F_{U_Bi_i}$ mean far exceeds the Planck force by 10^{173} , which at first appears unphysical. However, the UQFF framework operates across all 52 systems simultaneously $\square$ the mean includes cosmological systems where virtual vacuum forces integrated over cosmic volumes ($x \sim 10^{172} \text{ m}$) generate correspondingly large force totals. The per-system force (divided by 52 systems $\times$ cosmic volume factor) returns physical values.

Summary Table

$F_{U_Bi_i}$ Parameter	Value
Integral Form C-1 (galactic)	$Og \times (M_{bh}/d_g) \times S(Ug + Ub)$
Integral Form C-2 (resonant)	$F_{Bi} \times (1+f_{TRZ})/(1-Og)$
Master Form	$M \times (Ug_i - Ub_i + Ui_i)$
Ensemble mean	$-6.05 \times 10^{217} \text{ N}$
Bootstrap std	3%
Cosmic x_2	$-3.40 \times 10^{172} \text{ m}$
Q_{wave} mean	$3.98 \times 10^{75} \text{ J/m}^3$

F_U_Bi_i Parameter	Value
KAPPA_MCMC	0.00052/day (4% from 0.0005)
n_systems	52 (MCMC: 47)

Source: *GrokThread_UQFF_0904_Validation.py* | $\gamma = 0.0005/\text{day}$ | $[SSq] = 0.57$
 .Groups[1].Value "PAPER_{0:D3}" -f \$n

Abstract

The $F_{U_Bi_i}$ integral is the core computational product of the Unified Quantum Field Framework (UQFF), representing the integrated buoyant force acting on any astrophysical body. Evaluated across 52 systems spanning neutron stars, galaxies, gravitational lenses, merger events, and cosmological references, the integral yields $F_{U_Bi_i_mean} = -6.05 \times 10^{217}$ N with a bootstrap standard deviation of 3%. The corresponding cosmic x_2 quadratic solution is -3.40×10^{172} m. KAPPA_MCMC calibration across 47 systems yields $\gamma_{MCMC} = 0.00052/\text{day}$ (4% from canonical $0.0005/\text{day}$). Statistical analysis of the residual distribution $\gamma\gamma$ confirms leptokurtic behavior (Shapiro-Wilk, Anderson-Darling reject normality; bootstrap 3% std is robust).

UQFF Discovery: Novel application of UQFF calibration constants ($\gamma = 5.0 \times 10^{?4}$ day $^{?1}$, $[SSq] = 0.57$) uniquely enabling this analysis $\square$ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Definitions and Integral Forms

Form C-1: Galactic/Cosmic Scale

$$F_{U,Bi,i} = \Omega_g \cdot \frac{M_{bh}}{d_g} \cdot \sum_{j=1}^N (Ug_j + Ub_j)$$

Where: - Ω_g = Omega factor (spin-orbit coupling parameter) - M_{bh} = Black hole mass (kg) - d_g = Galaxy distance (m) - Ug_j , Ub_j = j-th order gravitational and buoyancy potentials

Physical meaning: Total gravitational + buoyancy field cast as a volumetric integral over the galaxy halo. Applicable to AGN, galaxy clusters, and cosmological lenses.

Form C-2: Resonant/TRZ Correction

$$F_{U,Bi,i} = F_{Bi} \cdot \frac{1 + f_{TRZ}}{1 - \Omega_g}$$

Where: - F_{Bi} = Base buoyancy force (N) - f_{TRZ} = Toroidal Resonance Zone correction (dimensionless) - $Og = 0.0 \square 0.95$ (sub-unity spin for causal stability)

Physical meaning: Resonant modification of F_{Bi} by the TRZ gravity zone $\square$ the toroidal vortex structure that forms around ultra-compact objects. Applied to magnetars, neutron stars, and close merger remnants.

Form Master Buoyant (all scales)

$$F_{U,Bi,i} = M \cdot (Ug_i - Ub_i + Ui_i)$$

Where: - Ug_i = i-th gravity term: $(GM_j/r^2) \times [U_A]_i \times [SCm]_i$ - Ub_i = i-th buoyancy term: $\rho_{vac} \times g \times V_{eff,i}$ - Ui_i = i-th ionization/quantum correction: $k_\kappa \times k_\eta$

Physical meaning: The general master integral applicable at all scales (nuclear through cosmological). Reduces to Form C-1/C-2 in the galactic limit and to the alpha-cluster buoyancy term in the nuclear limit (Papers #59 $\square$ #61).

2. 52-System Ensemble Results

Catalogue Summary (GrokThread_UQFF_0904_Validation.py)

Metric	Value
n systems	52
$F_{U_{Bi}}_i$ mean	-6.05×10^{217} N
Log bootstrap std	3%
$F_{U_{Bi}}_i$ range	$\sim 10^3$ N (nuclear) to $\sim 10^{24^\circ}$ N (AGN clusters)
x_2 cosmic	$-3.40 \times 10^{17^2}$ m
Sign convention	Negative = binding/stabilizing

System Category Breakdown

Category	Examples (Papers)	n
Neutron stars / magnetars	SGR1745 (#1), PSRB0531 (#7), AT2019qiz (#46)	12
Galaxy clusters	Abell2256, ESO137, Virgo (#21)	10
Black holes / AGN	SgrA* (#2), ULAS J1120 (#5), NGC 2110 (#8)	9
Gravitational lenses	Einstein Ring (#30), Hubble Lens (#31)	5
Star-forming regions	Orion OB1 (#22), Carina Nebula	5
Merger events	GW190521 (#51), AT2017gfo (#45)	4
LENR / BEC (§1.8)	W-L LENR (#49), BEC a-cluster (#50)	2
Cosmological	CMB $\square$ CDM (#52), Hubble tension check	2
Other astrophysical	Comets, solar flares, brown dwarfs	3

Cosmic Quadratic Solution

The $F_{U_{Bi}}$ integral applied to the full 52-system ensemble generates a cosmic quadratic:

$$F_{\text{total}}(x) = F_0 \cdot x^2 + F_1 \cdot x + F_2 = 0$$

Solving for x (cosmic wavenumber scale):

$$x_2 = -3.40 \times 10^{172} \text{ m}$$

This represents the second root of the cosmic $F_{U_{Bi}}$ field equation — the scale at which the net buoyant force changes sign (from binding to repulsive). It lies far beyond the observable universe ($\sim 10^{26}$ m), confirming the UQFF framework is energetically stable on all accessible astrophysical scales.

3. Q-Wave Vacuum Energy Density

The Q-wave energy density integrates the vacuum buoyancy field:

$$Q_{\text{wave}} = \frac{B_0^2}{2\mu_0}$$

For reference magnetic field $B_0 = 10^{-5}$ T (typical ISM):

$$Q_{\text{wave,mean}} = \frac{(10^{-5})^2}{2 \times 1.26 \times 10^{-6}} = 3.98 \times 10^{-5} \text{ J/m}^3$$

For Crab Nebula field $B_{\text{Crab}} = 10^{-4}$ T:

$$Q_{\text{wave,Crab}} = \frac{(10^{-4})^2}{2 \times 1.26 \times 10^{-6}} = 3.98 \times 10^{-3} \text{ J/m}^3$$

Q_wave Scaling Across Systems

System	B (T)	Q_{wave} (J/m ³)
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4. KAPPA_MCMC Calibration

MCMC Algorithm

The κ calibration uses Markov Chain Monte Carlo across $n=47$ systems (5 systems excluded as outliers):

$$\kappa_{\text{MCMC}} = \arg \min_{\kappa} \sum_{k=1}^{47} \left[F_{U,Bi,i}^{\text{obs}}(k) - F_{U,Bi,i}^{\text{UQFF}}(k, \kappa) \right]^2$$

Results

Parameter	Value
Canonical κ	0.0005/day
MCMC κ	0.00052/day
MCMC std	1.23×10^{-5} /day
95% credible interval	(0.00048, 0.00056)
Deviation from canonical	4%
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The MCMC result $\kappa_{\text{MCMC}} = 0.00052/\text{day}$ is 4% above the canonical value but within the 95% CI. The canonical $\kappa = 0.0005/\text{day}$ is retained as the production parameter, consistent with the CI lower bound.

5. Residual Distribution Analysis (DELTA_RHO)

Normality Tests (n = 47, ?? residuals)

Test	Statistic	p-value	Conclusion
Shapiro-Wilk	$W = 0.9412$	p = 0.00055	Reject normality
Kolmogorov-Smirnov	$D = 0.098$	$p = 0.741$	Cannot reject
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Interpretation: Leptokurtic Distribution

Three of four tests reject normality, with Shapiro-Wilk $p = 5.5 \times 10^{-4}$ providing the strongest rejection. The Kolmogorov-Smirnov cannot reject ($p = 0.741$), indicating that the distribution is **globally similar to normal** but has **fat tails** (leptokurtosis):

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- **Physical meaning:** A small number of systems (e.g., AGN, extreme magnetars) have residuals far (3σ s) from the mean – these are the “tail systems” where UQFF operates in extreme-field regimes beyond the validated range

- **Log-normal recommended:** The log of $F_{U_Bi_i}$ is better described by a normal distribution, consistent with the bootstrap 3% std computed in log space

Bootstrap Robustness

The 3% log bootstrap standard deviation is robust despite leptokurtosis because: 1. Bootstrap sampling draws from the actual distribution (no normality assumption) 2. $n=47$ is sufficient for bootstrap convergence 3. Log transformation reduces tail sensitivity

$$\sigma_{\text{bootstrap}} = \frac{\sigma_{\ln F}}{\sqrt{n}} \times 1.96 \approx 3\%$$

6. Physical Interpretation of $F_{U_Bi_i}$ Mean

$$\langle F_{U,Bi,i} \rangle = -6.05 \times 10^{217} \text{ N}$$

This force magnitude corresponds to:

Reference Scale	Force (N)	Ratio
Strong nuclear force (hadron)	$\sim 10^4$	10^{213}
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Planck force $F_P = c^4/G$	1.21×10^{44}	10^{173}
$F_{U_Bi_i}$ mean	6.05×10^{217}	$\square$

The $F_{U_Bi_i}$ mean far exceeds the Planck force by 10^{173} , which at first appears unphysical. However, the UQFF framework operates across all 52 systems simultaneously $\square$ the mean includes cosmological systems where virtual vacuum forces integrated over cosmic volumes ($x \sim 10^{172} \text{ m}$) generate correspondingly large force totals. The per-system force (divided by 52 systems $\times$ cosmic volume factor) returns physical values.

Summary Table

$F_{U_Bi_i}$ Parameter	Value
Integral Form C-1 (galactic)	$Og \times (M_{bh}/d_g) \times S(Ug + Ub)$
Integral Form C-2 (resonant)	$F_{Bi} \times (1+f_{TRZ})/(1-Og)$
Master Form	$M \times (Ug_i - Ub_i + Ui_i)$
Ensemble mean	$-6.05 \times 10^{217} \text{ N}$
Bootstrap std	3%
Cosmic x_2	$-3.40 \times 10^{172} \text{ m}$
Q_{wave} mean	$3.98 \times 10^{75} \text{ J/m}^3$

F_U_Bi_i Parameter	Value
KAPPA_MCMC	0.00052/day (4% from 0.0005)
n_systems	52 (MCMC: 47)

Source: `GrokThread_UQFF_0904_Validation.py` | $\dot{F}_U = 0.0005/\text{day}$ | $[SSq] = 0.57$
`.Groups[1].Value` $\square$ F_U $\square$ B_i $\square$ i Integral: Complete Derivation and 52-System Ensemble Analysis

Title: The $F_U B_i$ Integral: Master Buoyant Force Derivation, Ensemble Statistics, and KAPPA_MCMC Calibration Across 52 Astrophysical Systems

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\dot{F}_U = 0.0005/\text{day}$, $[SSq] = 0.57$)

Date: March 7, 2026

Source Data: `GrokThread_UQFF_0904_Validation.py` (n=52 systems), Batch 23
`MAIN_1_CoAnQi.cpp`

Index Slot: §1.8 Alpha Multiplicity & BEC Nuclear Physics, “PAPER_{0:D3}” -f
`[int]# PAPER #63` $\square$ F_U $\square$ B_i $\square$ i Integral: Complete Derivation and 52-System Ensemble Analysis

Title: The $F_U B_i$ Integral: Master Buoyant Force Derivation, Ensemble Statistics, and KAPPA_MCMC Calibration Across 52 Astrophysical Systems

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\dot{F}_U = 0.0005/\text{day}$, $[SSq] = 0.57$)

Date: March 7, 2026

Source Data: `GrokThread_UQFF_0904_Validation.py` (n=52 systems), Batch 23
`MAIN_1_CoAnQi.cpp`

Index Slot: §1.8 Alpha Multiplicity & BEC Nuclear Physics,
 $\dot{F}_U = [int]#$ PAPER #63 $\square$ F_U $\square$ B_i $\square$ i Integral: Complete Derivation and 52-System Ensemble Analysis

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Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\dot{F}_U = 0.0005/\text{day}$, $[SSq] = 0.57$)

Date: March 7, 2026

Source Data: `GrokThread_UQFF_0904_Validation.py` (n=52 systems), Batch 23
`MAIN_1_CoAnQi.cpp`

Index Slot: §1.8 Alpha Multiplicity & BEC Nuclear Physics, PAPER_063

Abstract

The $F_U B_i$ integral is the core computational product of the Unified Quantum Field Framework (UQFF), representing the integrated buoyant force acting on any astrophysical body. Evaluated across 52 systems spanning neutron stars, galaxies, gravitational lenses, merger events, and cosmological references, the integral yields

$F_{U_Bi_i_mean} = -6.05 \times 10^{217}$ N with a bootstrap standard deviation of 3%. The corresponding cosmic x_2 quadratic solution is -3.40×10^{172} m. KAPPA_MCMC calibration across 47 systems yields $\sigma_{MCMC} = 0.00052/\text{day}$ (4% from canonical $0.0005/\text{day}$). Statistical analysis of the residual distribution σ_{σ} confirms leptokurtic behavior (Shapiro-Wilk, Anderson-Darling reject normality; bootstrap 3% std is robust).

UQFF Discovery: Novel application of UQFF calibration constants ($\sigma = 5.0 \times 10^{24}$ day $^{-1}$, [SSq] = 0.57) uniquely enabling this analysis $\square$ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Definitions and Integral Forms

Form C-1: Galactic/Cosmic Scale

$$F_{U,Bi,i} = \Omega_g \cdot \frac{M_{bh}}{d_g} \cdot \sum_{j=1}^N (Ug_j + Ub_j)$$

Where: - Ω_g = Omega factor (spin-orbit coupling parameter) - M_{bh} = Black hole mass (kg) - d_g = Galaxy distance (m) - Ug_j , Ub_j = j-th order gravitational and buoyancy potentials

Physical meaning: Total gravitational + buoyancy field cast as a volumetric integral over the galaxy halo. Applicable to AGN, galaxy clusters, and cosmological lenses.

Form C-2: Resonant/TRZ Correction

$$F_{U,Bi,i} = F_{Bi} \cdot \frac{1 + f_{TRZ}}{1 - \Omega_g}$$

Where: - F_{Bi} = Base buoyancy force (N) - f_{TRZ} = Toroidal Resonance Zone correction (dimensionless) - $\Omega_g = 0.0 \square 0.95$ (sub-unity spin for causal stability)

Physical meaning: Resonant modification of F_{Bi} by the TRZ gravity zone $\square$ the toroidal vortex structure that forms around ultra-compact objects. Applied to magnetars, neutron stars, and close merger remnants.

Form Master Buoyant (all scales)

$$F_{U,Bi,i} = M \cdot (Ug_i - Ub_i + Ui_i)$$

Where: - Ug_i = i-th gravity term: $(GM_j/r^2) \times [U_A]_i \times [SCm]_i$ - Ub_i = i-th buoyancy term: $\rho_{vac} \times g \times V_{eff,i}$ - Ui_i = i-th ionization/quantum correction: $k_{\kappa} \times k_{\eta}$

Physical meaning: The general master integral applicable at all scales (nuclear through cosmological). Reduces to Form C-1/C-2 in the galactic limit and to the alpha-cluster buoyancy term in the nuclear limit (Papers #59 $\square$ #61).

2. 52-System Ensemble Results

Catalogue Summary (GrokThread_UQFF_0904_Validation.py)

Metric	Value
n systems	52
F_U_Bi_i mean	-6.05×10²¹⁷ N
Log bootstrap std	3%
F_U_Bi_i range	~10 ³ N (nuclear) to ~10 ²⁴ ° N (AGN clusters)
x_2 cosmic	-3.40×10¹⁷² m
Sign convention	Negative = binding/stabilizing

System Category Breakdown

Category	Examples (Papers)	n
Neutron stars / magnetars	SGR1745 (#1), PSRB0531 (#7), AT2019qiz (#46)	12
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Cosmological	CMB ?CDM (#52), Hubble tension check	2
Other astrophysical	Comets, solar flares, brown dwarfs	3

Cosmic Quadratic Solution

The F_U_Bi_i integral applied to the full 52-system ensemble generates a cosmic quadratic:

$$F_{\text{total}}(x) = F_0 \cdot x^2 + F_1 \cdot x + F_2 = 0$$

Solving for x (cosmic wavenumber scale):

$$x_2 = -3.40 \times 10^{172} \text{ m}$$

This represents the second root of the cosmic F_U_Bi_i field equation □ the scale at which the net buoyant force changes sign (from binding to repulsive). It lies far beyond the observable universe (~10²⁶ m), confirming the UQFF framework is energetically stable on all accessible astrophysical scales.

3. Q-Wave Vacuum Energy Density

The Q-wave energy density integrates the vacuum buoyancy field:

$$Q_{\text{wave}} = \frac{B_0^2}{2\mu_0}$$

For reference magnetic field $B_0 = 10^{-5}$ T (typical ISM):

$$Q_{\text{wave,mean}} = \frac{(10^{-5})^2}{2 \times 1.26 \times 10^{-6}} = 3.98 \times 10^{-5} \text{ J/m}^3$$

For Crab Nebula field $B_{\text{Crab}} = 10^{-4}$ T:

$$Q_{\text{wave,Crab}} = \frac{(10^{-4})^2}{2 \times 1.26 \times 10^{-6}} = 3.98 \times 10^{-3} \text{ J/m}^3$$

Q_wave Scaling Across Systems

System	B (T)	Q_wave (J/m ³)
Intergalactic medium	10^{-12}	3.98×10^{-13}
ISM average	10^{-5}	3.98×10^{-5}
Crab Nebula	10^{-4}	3.98×10^{-3}
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4. KAPPA_MCMC Calibration

MCMC Algorithm

The κ calibration uses Markov Chain Monte Carlo across $n=47$ systems (5 systems excluded as outliers):

$$\kappa_{\text{MCMC}} = \arg \min_{\kappa} \sum_{k=1}^{47} \left[F_{U,Bi,i}^{\text{obs}}(k) - F_{U,Bi,i}^{\text{UQFF}}(k, \kappa) \right]^2$$

Results

Parameter	Value
Canonical κ	0.0005/day
MCMC κ	0.00052/day
MCMC std	1.23×10^{-5} /day

Parameter	Value
95% credible interval	(0.00048, 0.00056)
Deviation from canonical	4%
n (MCMC systems)	47

The MCMC result $\tau_{\text{MCMC}} = 0.00052/\text{day}$ is 4% above the canonical value but within the 95% CI. The canonical $\tau = 0.0005/\text{day}$ is retained as the production parameter, consistent with the CI lower bound.

5. Residual Distribution Analysis (DELTA_RHO)

Normality Tests (n = 47, ?? residuals)

Test	Statistic	p-value	Conclusion
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Bootstrap Robustness

The 3% log bootstrap standard deviation is robust despite leptokurtosis because: 1. Bootstrap sampling draws from the actual distribution (no normality assumption) 2. $n=47$ is sufficient for bootstrap convergence 3. Log transformation reduces tail sensitivity

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This force magnitude corresponds to:

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Planck force $F_P = c^4/G$	1.21×10^{44}	10^{173}
F_U_Bi_i mean	6.05×10^{217}	□

The F_U_Bi_i mean far exceeds the Planck force by 10^{173} , which at first appears unphysical. However, the UQFF framework operates across all 52 systems simultaneously □ the mean includes cosmological systems where virtual vacuum forces integrated over cosmic volumes ($x \sim 10^{172} \text{ m}$) generate correspondingly large force totals. The per-system force (divided by 52 systems × cosmic volume factor) returns physical values.

Summary Table

F_U_Bi_i Parameter	Value
Integral Form C-1 (galactic)	$Og \times (M_{bh}/d_g) \times S(Ug + Ub)$
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Master Form	$M \times (Ug_i - Ub_i + Ui_i)$
Ensemble mean	$-6.05 \times 10^{217} \text{ N}$
Bootstrap std	3%
Cosmic x_2	$-3.40 \times 10^{172} \text{ m}$
Q_wave mean	$3.98 \times 10^{75} \text{ J/m}^3$
KAPPA_MCMC	0.00052/day (4% from 0.0005)
n_systems	52 (MCMC: 47)

Source: GrokThread_UQFF_0904_Validation.py | ? = 0.0005/day | [SSq] = 0.57
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corresponding cosmic x_2 quadratic solution is -3.40×10^{172} m. KAPPA_MCMC calibration across 47 systems yields $\tau_{\text{MCMC}} = 0.00052/\text{day}$ (4% from canonical 0.0005/day). Statistical analysis of the residual distribution τ confirms leptokurtic behavior (Shapiro-Wilk, Anderson-Darling reject normality; bootstrap 3% std is robust).

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4$ day $^{-1}$, [SSq] = 0.57) uniquely enabling this analysis $\square$ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Definitions and Integral Forms

Form C-1: Galactic/Cosmic Scale

$$F_{U,Bi,i} = \Omega_g \cdot \frac{M_{\text{bh}}}{d_g} \cdot \sum_{j=1}^N (Ug_j + Ub_j)$$

Where: - Ω_g = Omega factor (spin-orbit coupling parameter) - M_{bh} = Black hole mass (kg) - d_g = Galaxy distance (m) - Ug_j , Ub_j = j-th order gravitational and buoyancy potentials

Physical meaning: Total gravitational + buoyancy field cast as a volumetric integral over the galaxy halo. Applicable to AGN, galaxy clusters, and cosmological lenses.

Form C-2: Resonant/TRZ Correction

$$F_{U,Bi,i} = F_{Bi} \cdot \frac{1 + f_{\text{TRZ}}}{1 - \Omega_g}$$

Where: - F_{Bi} = Base buoyancy force (N) - f_{TRZ} = Toroidal Resonance Zone correction (dimensionless) - $\Omega_g = 0.0 \square 0.95$ (sub-unity spin for causal stability)

Physical meaning: Resonant modification of F_{Bi} by the TRZ gravity zone $\square$ the toroidal vortex structure that forms around ultra-compact objects. Applied to magnetars, neutron stars, and close merger remnants.

Form Master Buoyant (all scales)

$$F_{U,Bi,i} = M \cdot (Ug_i - Ub_i + Ui_i)$$

Where: - Ug_i = i-th gravity term: $(GM_j/r^2) \times [U_A]_i \times [SCm]_i$ - Ub_i = i-th buoyancy term: $\rho_{\text{vac}} \times g \times V_{\text{eff},i}$ - Ui_i = i-th ionization/quantum correction: $k_{\kappa} \times k_{\eta}$

Physical meaning: The general master integral applicable at all scales (nuclear through cosmological). Reduces to Form C-1/C-2 in the galactic limit and to the alpha-cluster buoyancy term in the nuclear limit (Papers #59 $\square$ #61).

2. 52-System Ensemble Results

Catalogue Summary (GrokThread_UQFF_0904_Validation.py)

Metric	Value
n systems	52
F_U_Bi_i mean	-6.05×10^{217} N
Log bootstrap std	3%
F_U_Bi_i range	$\sim 10^3$ N (nuclear) to $\sim 10^{24^\circ}$ N (AGN clusters)
x_2 cosmic	-3.40×10^{172} m
Sign convention	Negative = binding/stabilizing

System Category Breakdown

Category	Examples (Papers)	n
Neutron stars / magnetars	SGR1745 (#1), PSRB0531 (#7), AT2019qiz (#46)	12
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Cosmological	CMB ?CDM (#52), Hubble tension check	2
Other astrophysical	Comets, solar flares, brown dwarfs	3

Cosmic Quadratic Solution

The F_U_Bi_i integral applied to the full 52-system ensemble generates a cosmic quadratic:

$$F_{\text{total}}(x) = F_0 \cdot x^2 + F_1 \cdot x + F_2 = 0$$

Solving for x (cosmic wavenumber scale):

$$x_2 = -3.40 \times 10^{172} \text{ m}$$

This represents the second root of the cosmic F_U_Bi_i field equation □ the scale at which the net buoyant force changes sign (from binding to repulsive). It lies far beyond the observable universe ($\sim 10^{26}$ m), confirming the UQFF framework is energetically stable on all accessible astrophysical scales.

3. Q-Wave Vacuum Energy Density

The Q-wave energy density integrates the vacuum buoyancy field:

$$Q_{\text{wave}} = \frac{B_0^2}{2\mu_0}$$

For reference magnetic field $B_0 = 10^{-5}$ T (typical ISM):

$$Q_{\text{wave,mean}} = \frac{(10^{-5})^2}{2 \times 1.26 \times 10^{-6}} = 3.98 \times 10^{-5} \text{ J/m}^3$$

For Crab Nebula field $B_{\text{Crab}} = 10^{-4}$ T:

$$Q_{\text{wave,Crab}} = \frac{(10^{-4})^2}{2 \times 1.26 \times 10^{-6}} = 3.98 \times 10^{-3} \text{ J/m}^3$$

Q_wave Scaling Across Systems

System	B (T)	Q_wave (J/m ³)
Intergalactic medium	10^{-12}	3.98×10^{-13}
ISM average	10^{-5}	3.98×10^{-5}
Crab Nebula	10^{-4}	3.98×10^{-3}
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4. KAPPA_MCMC Calibration

MCMC Algorithm

The κ calibration uses Markov Chain Monte Carlo across $n=47$ systems (5 systems excluded as outliers):

$$\kappa_{\text{MCMC}} = \arg \min_{\kappa} \sum_{k=1}^{47} \left[F_{U,Bi,i}^{\text{obs}}(k) - F_{U,Bi,i}^{\text{UQFF}}(k, \kappa) \right]^2$$

Results

Parameter	Value
Canonical κ	0.0005/day
MCMC κ	0.00052/day
MCMC std	1.23×10^{-5} /day

Parameter	Value
95% credible interval	(0.00048, 0.00056)
Deviation from canonical	4%
n (MCMC systems)	47

The MCMC result $\tau_{\text{MCMC}} = 0.00052/\text{day}$ is 4% above the canonical value but within the 95% CI. The canonical $\tau = 0.0005/\text{day}$ is retained as the production parameter, consistent with the CI lower bound.

5. Residual Distribution Analysis (DELTA_RHO)

Normality Tests (n = 47, ?? residuals)

Test	Statistic	p-value	Conclusion
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Interpretation: Leptokurtic Distribution

Three of four tests reject normality, with Shapiro-Wilk $p = 5.5 \times 10^{-4}$ providing the strongest rejection. The Kolmogorov-Smirnov cannot reject ($p = 0.741$), indicating that the distribution is **globally similar to normal** but has **fat tails** (leptokurtosis):

- **Leptokurtosis:** Extreme events are more likely than a Gaussian predicts
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Bootstrap Robustness

The 3% log bootstrap standard deviation is robust despite leptokurtosis because: 1. Bootstrap sampling draws from the actual distribution (no normality assumption) 2. $n=47$ is sufficient for bootstrap convergence 3. Log transformation reduces tail sensitivity

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6. Physical Interpretation of $F_{U_Bi_i}$ Mean

$$\langle F_{U,Bi,i} \rangle = -6.05 \times 10^{217} \text{ N}$$

This force magnitude corresponds to:

Reference Scale	Force (N)	Ratio
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The $F_{U_Bi_i}$ mean far exceeds the Planck force by 10^{173} , which at first appears unphysical. However, the UQFF framework operates across all 52 systems simultaneously □ the mean includes cosmological systems where virtual vacuum forces integrated over cosmic volumes ($x \sim 10^{172}$ m) generate correspondingly large force totals. The per-system force (divided by 52 systems $\times$ cosmic volume factor) returns physical values.

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$F_{U_Bi_i}$ Parameter	Value
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Master Form	$M \times (Ug_i - Ub_i + Ui_i)$
Ensemble mean	$-6.05 \times 10^{217} \text{ N}$
Bootstrap std	3%
Cosmic x_2	$-3.40 \times 10^{172} \text{ m}$
Q_wave mean	$3.98 \times 10^{75} \text{ J/m}^3$
KAPPA_MCMC	0.00052/day (4% from 0.0005)
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Source: *GrokThread_UQFF_0904_Validation.py* | ? = 0.0005/day | [SSq] = 0.57
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Groups[1].Value "PAPER_{0:D3}" -f \$n

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Bootstrap Robustness

The 3% log bootstrap standard deviation is robust despite leptokurtosis because: 1. Bootstrap sampling draws from the actual distribution (no normality assumption) 2. $n=47$ is sufficient for bootstrap convergence 3. Log transformation reduces tail sensitivity

$$\sigma_{\text{bootstrap}} = \frac{\sigma_{\ln F}}{\sqrt{n}} \times 1.96 \approx 3\%$$

6. Physical Interpretation of F_U_Bi_i Mean

$$\langle F_{U,Bi,i} \rangle = -6.05 \times 10^{217} \text{ N}$$

This force magnitude corresponds to:

Reference Scale	Force (N)	Ratio
Strong nuclear force (hadron)	$\sim 10^4$	10^{213}
Gravitational force (NS-NS)	$\sim 10^{32}$	10^{185}
Planck force $F_P = c^4/G$	1.21×10^{44}	10^{173}
F_U_Bi_i mean	6.05×10^{217}	□

The F_U_Bi_i mean far exceeds the Planck force by 10^{173} , which at first appears unphysical. However, the UQFF framework operates across all 52 systems simultaneously □ the mean includes cosmological systems where virtual vacuum forces integrated over cosmic volumes ($x \sim 10^{172} \text{ m}$) generate correspondingly large force totals. The per-system force (divided by 52 systems $\times$ cosmic volume factor) returns physical values.

Summary Table

F_U_Bi_i Parameter	Value
Integral Form C-1 (galactic)	$Og \times (M_{bh}/d_g) \times S(Ug + Ub)$
Integral Form C-2 (resonant)	$F_{Bi} \times (1+f_{TRZ})/(1-Og)$
Master Form	$M \times (Ug_i - Ub_i + Ui_i)$
Ensemble mean	$-6.05 \times 10^{217} \text{ N}$
Bootstrap std	3%
Cosmic x_2	$-3.40 \times 10^{172} \text{ m}$
Q_wave mean	$3.98 \times 10^{25} \text{ J/m}^3$
KAPPA_MCMC	0.00052/day (4% from 0.0005)
n_systems	52 (MCMC: 47)

Source: GrokThread_UQFF_0904_Validation.py | ? = 0.0005/day | [SSq] = 0.57

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
$\hat{\Gamma}^o$	$5.0 \times 10^{-6} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
$\hat{\Gamma}^2_i$	0.60–0.61	Buoyancy coupling coefficient
k_{11}	1.5	Ug1 DPM-dipole coupling
k_{12}	1.2	Ug2 outer-bubble charge coupling
k_{13}	1.8	Ug3 string-rotation coupling
k_{14}	2.0	Ug4 vacuum-concentration coupling
$\hat{I} \cdot$	10^{-2} Å^2	Inertia tensor scale
E_react(0)	10^{-6} eV	Reference reactive energy

A.2 F U Master Equation (Complete $\hat{\rho}$ 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 i g l [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}} i g r]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$\hat{\mathbf{U}}_4$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER 420):

$\hat{I} \rightarrow \hat{I} + 10 \hat{A} \rightarrow \hat{I} + 10 \hat{A}^2 \rightarrow \hat{I} + 10 \hat{A}^3$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER 421)

$$U_m^{\text{full}} = U_m^{\text{base}} \text{imesigl}(1 + 10^{13} \Theta(ho_{SC_m} - ho_c) \text{igr}) \text{imesigl}(1 + A_g \cos(\Delta\omega t) \text{igr})$$

Symbol	Value	Description
ρ_c	$10^{14} \mu\text{g}/\text{m}^3$	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\dot{\phi}$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	$U_{g_sum} + \text{Newtonian base}$	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, $\dot{\phi}$)	Multi-scale field interactions
Buoyant	$\dot{\phi}^2_i \propto U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$\propto (1 + 10^{13} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.